

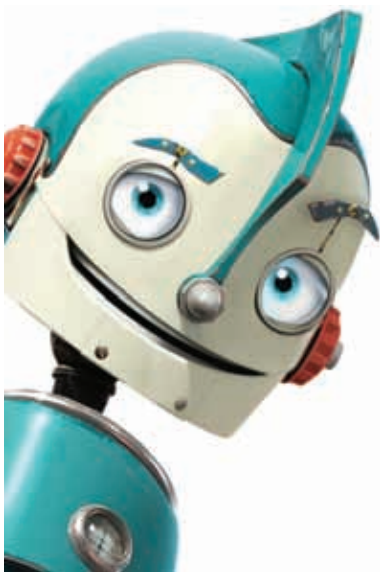
to the standard Turing Test, imagination in a visual sense would not be necessary of an AI since the entire test is conducted from a textual perspective. Even if you look at the Total Turing Test, it still needs the AI to only identify the visual cues it is given – its ability to imagine or “be creative” is never tested.

Yet, can we really have true AI without the ability to be creative and imaginative? One of the leading thinkers in artificial intelligence, Ray Kurzweil, certainly didn't think so and neither did Harold Cohen, the art professor at the University of California at San Diego. In the early 70s, Cohen created a new piece of AI that he called AARON, which was born out of the desire to answer a simple question: what are the minimum conditions under which a set of marks function as an image?

In an article for Stanford University, Cohen wrote: “On the simplest level it was not hard to propose a plausible answer: it required the spectator's belief that the marks had resulted from a purposeful human, or human-like, act – actions performed by a computer program possessing cognitive capabilities quite like the ones we use ourselves to make and to understand images. The earliest versions of AARON could do very little more than distinguish between figure and ground, closed forms

and open forms, and perform various simple manipulations on those structures. That might not have been enough, however, had AARON not performed, as humans do, in feedback mode. All its decisions about how to proceed with a drawing, from the lowest level of constructing a single line to higher-level issues of composition, were made by considering what it wanted to do in relation to what it had done already.”

What's remarkable about the program is that AARON is completely autonomous. Its intelligence lies not only in its ability to come up with art, but



A robot with emotion? Possible?